

Dallas_Crimes_Project

Adam Inga

2026-04-08

Disclaimer

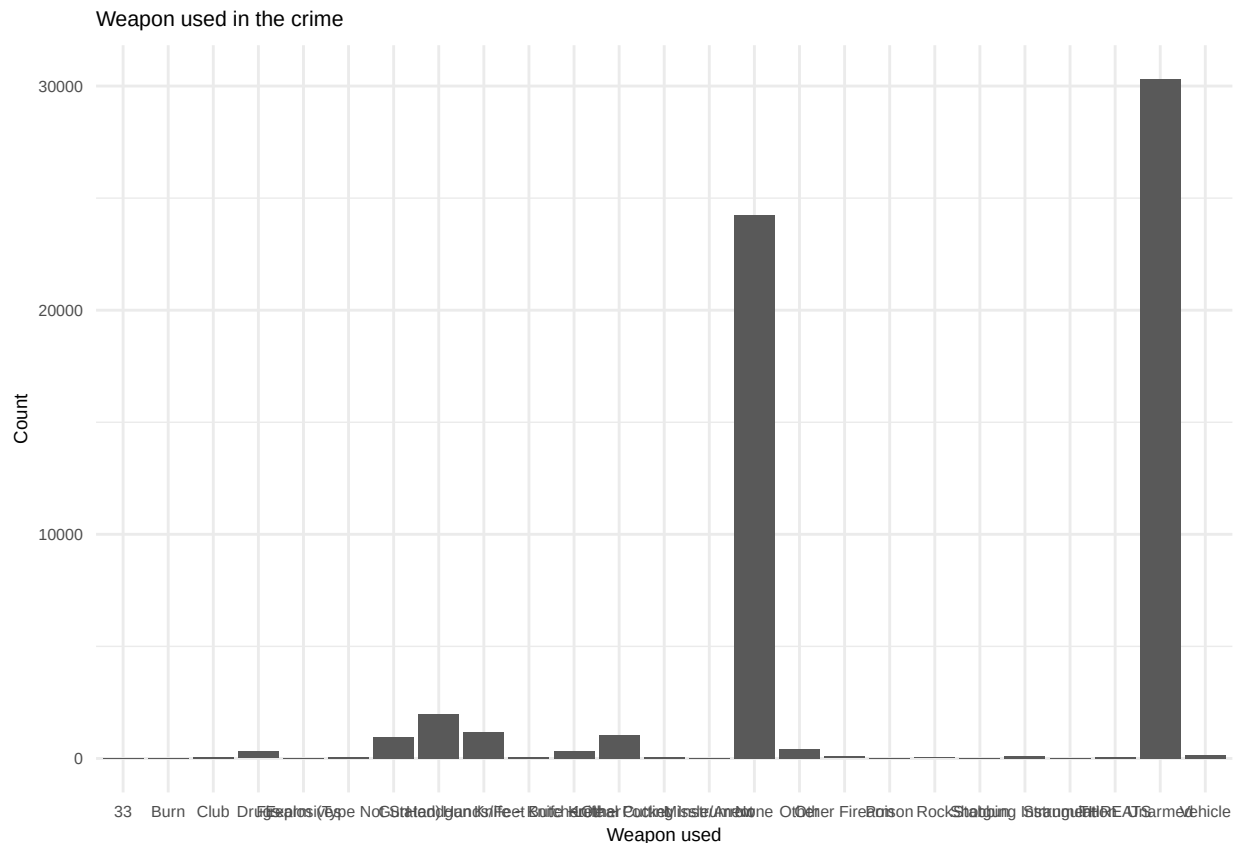
One important thing to note is that the data set from kaggle is 2014-2022. And due to the abundance of information in the Dallas crime open data set, we can cross reference them for good information from 2020-2022, and then use the Dallas crime open data set alone for some information from 2022-2025 and compare them.

Dataset 1

Data before cleaning

```
## open the data set from kaggle first
setwd("C:/Users/adamf/Downloads/Data (STATS)/archiveDallasCrimes")
dallasCrimesK <- read.csv("Police_Arrests.csv", na.strings = c("", "NA"))

# there are some outlines where some crime weapons are used so infrequently,
# they do not make a difference
ggplot(data = dallasCrimesK) + geom_bar(aes(x = Arrest.Weapon)) +
  theme_minimal() + theme(text = element_text(size = 7)) +
  labs(title = "Weapon used in the crime", x = "Weapon used", y = "Count")
```



The first data set that will be cleaned is the kaggle Dallas crimes data set. We see that there is a lot of variety in terms of the weapons used during crimes. Unfortunately there also seems to a lot of weapons whose counts are low they are practically invisible. The graph is so crowded even reducing the font size makes it hard to fit the categories and still be able to see what data is being presented.

In order to make it suitable to our other data set later on, after we filter by year, some of these counts will go down.

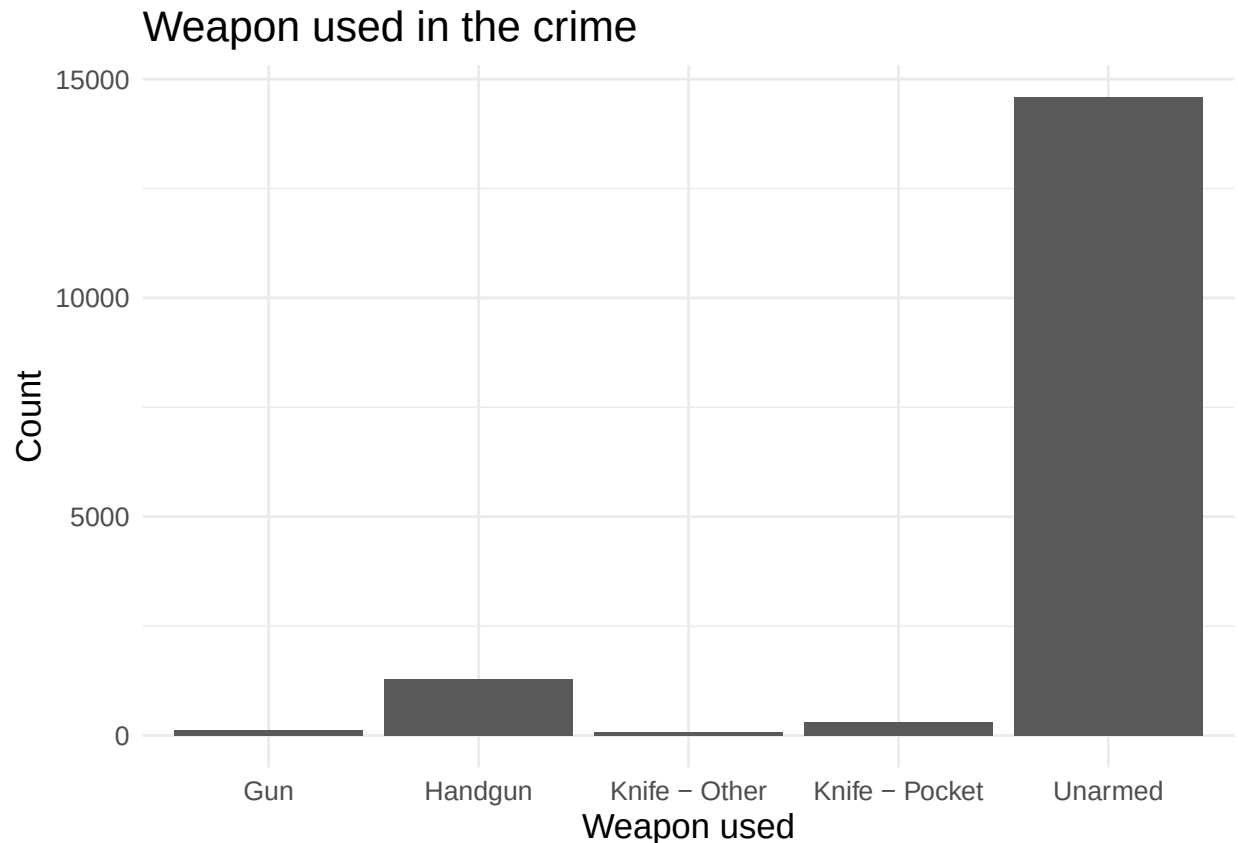
Data after cleaning

```
# here we will clean the data set

# we will first only use data from 2020 to the latest it goes to
dallasCrimesK2020 <- (dallasCrimesK[dallasCrimesK$Arrest.Year >= 2020, ])

# we will get rid of all the empty slots so we have good usable data
dallasCrimesK2020 <- dallasCrimesK2020 %>% group_by(Arrest.Weapon) %>%
  filter(n() > 50) %>% ungroup()

# graph the data again with bar graph
ggplot(data = dallasCrimesK2020) + geom_bar(aes(x = Arrest.Weapon)) +
  theme_minimal() + theme(text = element_text(size = 13)) +
  labs(title = "Weapon used in the crime", x = "Weapon used", y = "Count")
```



Now that we have finally filtered by year and taken out some unnecessary weapon categories, we see a more clear view of the data. Not only is it smaller but it is also easier to see and understand what categories are the most common.

Dataset 2

Before data cleaning

Now here we will combine the kaggle data set with the Dallas crime open data, we can do this using the incident number on each data set. From here we will determine what needs to be cleaned and what other modifications must be made in order to make it a usable data set.

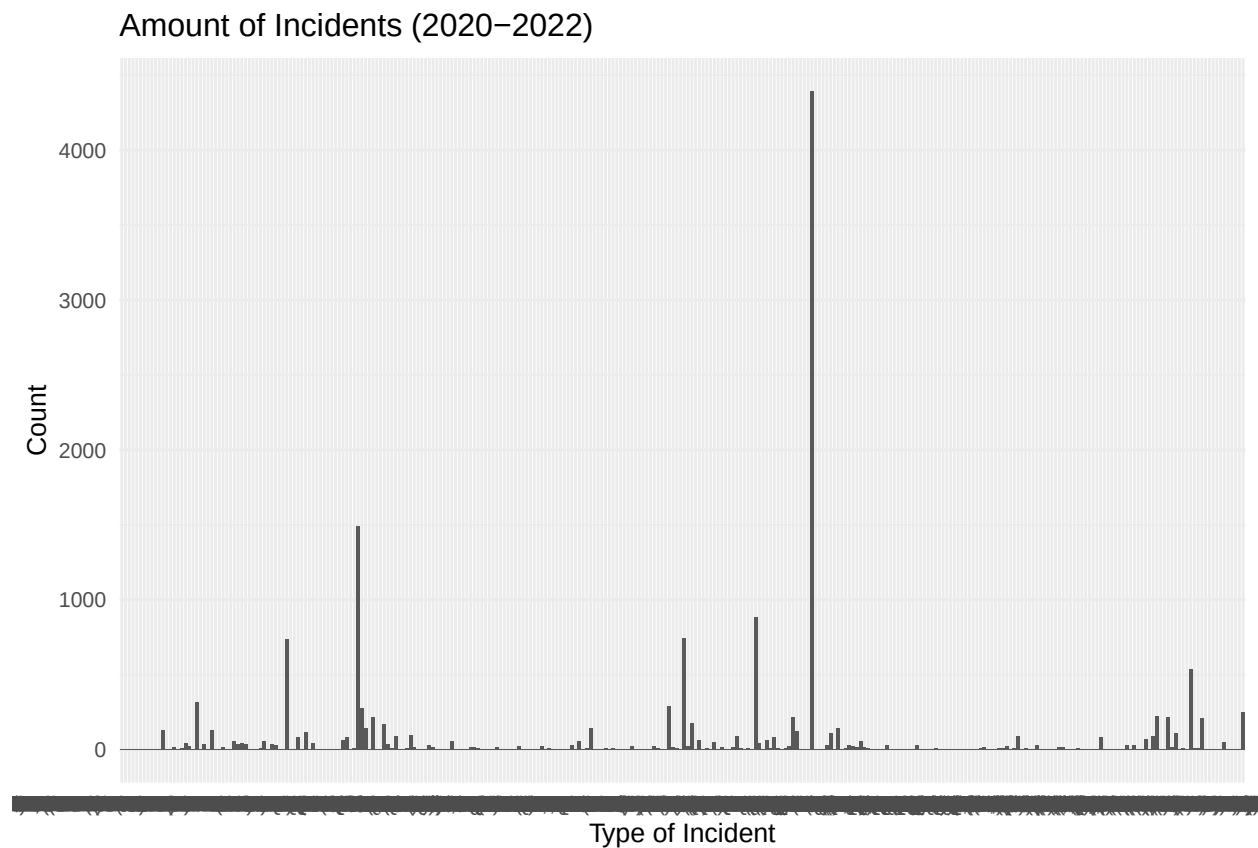
```
# This is the data from the Dallas police
setwd("C:/Users/adamf/Downloads/Data (STATS)/archiveDallasCrimes")
dallasCrimes <- read.csv("Police_Incidents_20260408(3).csv",
                        na.strings = c("", "NA"))

# keep only the first time an incident number appears
dallasCrimesK2020 <- dallasCrimesK2020 %>%
  distinct(IncidentNum, .keep_all = TRUE)

# keep only the first time an incident number appears
dallasCrimes <- dallasCrimes %>%
  distinct(Incident.Number.w.year, .keep_all = TRUE)
```

```
# we will keep incident number to cross reference with the main data set
result <- dallasCrimesK2020 %>% left_join(dallasCrimes,
                                           by = c("IncidentNum" = "Incident.Number.w.year"))

ggplot(data = result) + geom_bar(aes(x = Type.of.Incident)) +
  theme_minimal() + theme(text = element_text(size = 10)) +
  labs(title = "Amount of Incidents (2020-2022)",
       x = "Type of Incident", y = "Count")
```



The bar graph shows that there are a lot of categories but for most, its non existent as there aren't that many.

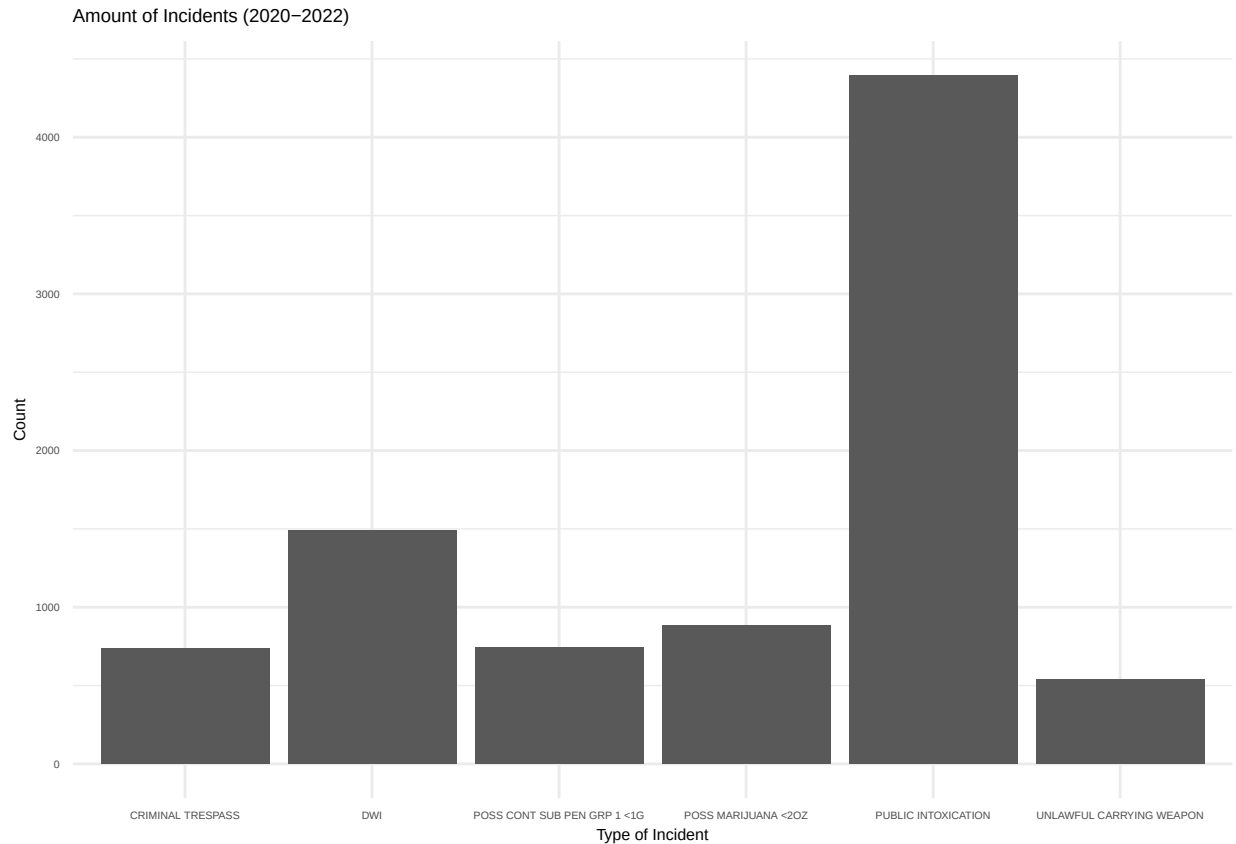
After data cleaning

```
# and here we get rid lf all the crimes that are with NA
resultClean <- result[!is.na(result$Type.of.Incident),, ]

# here we get all the crimes that happened at least 500 times
resultClean <-
  resultClean %>% group_by(Type.of.Incident) %>% filter(n() > 500) %>% ungroup()

# we graph the data again
ggplot(data = resultClean) + geom_bar(aes(x = Type.of.Incident)) +
  theme_minimal() + theme(text = element_text(size = 5.5)) +
```

```
labs(title = "Amount of Incidents (2020-2022)",
     x = "Type of Incident", y = "Count")
```



Dataset 3

Before cleaning

Here we will have the data from 2023 to present here and the data has already been filtered out before being downloaded to include 2023-present.

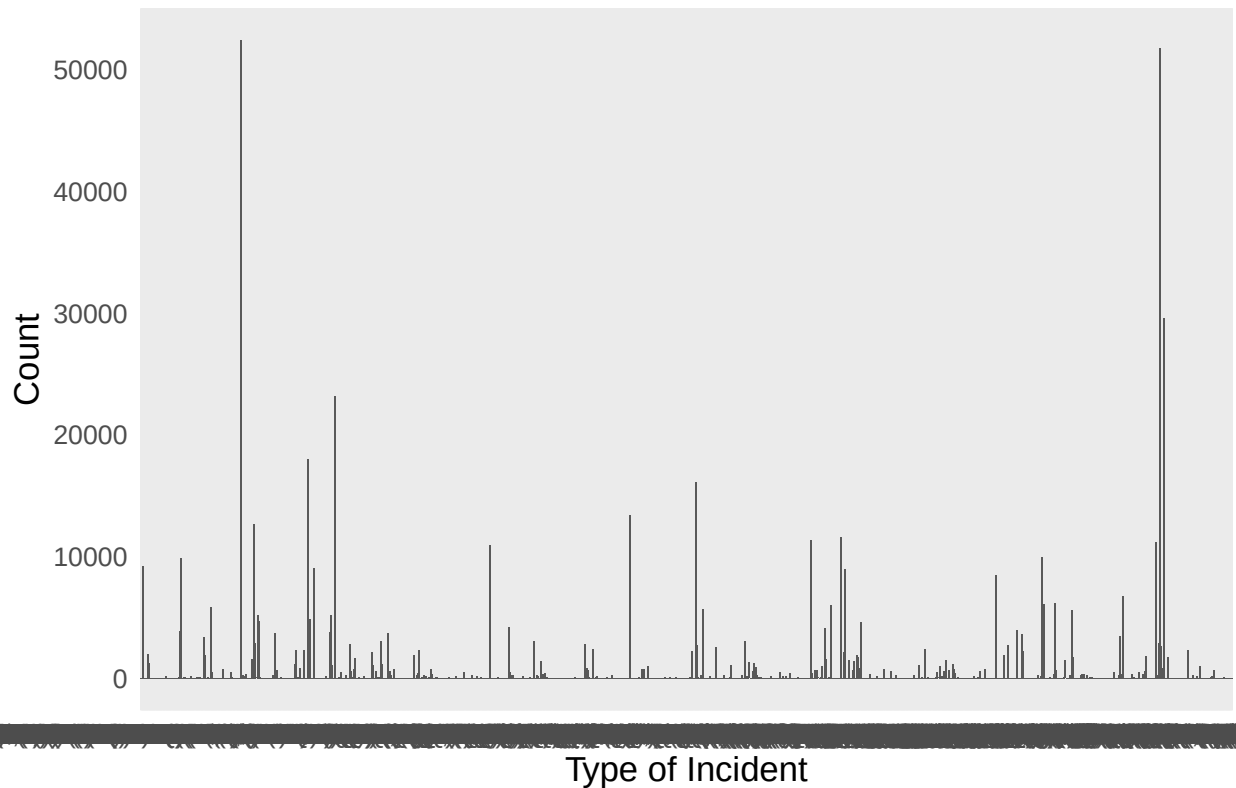
```
# This is the data from the Dallas police 2023 to present
setwd("C:/Users/adamf/Downloads/Data (STATS)/archiveDallasCrimes")
```

```
dallasCrimes2 <- read.csv("Police_Incidents_2023.csv", na.strings = c("", "NA"))
```

```
# here is the graph
```

```
ggplot(data = dallasCrimes2) + geom_bar(aes(x = Type.of.Incident)) +
  theme_minimal() + theme(text = element_text(size = 13)) +
  labs(title = "Amount of Incidents (2023-present)",
       x = "Type of Incident", y = "Count")
```

Amount of Incidents (2023–present)



The bar graph shows that there are a lot of categories but for most, it is non-existent as there aren't that many instances.

After data cleaning

```
# and here we get rid of all the crimes that are with NA
dallasCrimes2clean <- dallasCrimes2[!is.na(dallasCrimes2$Type.of.Incident), ]

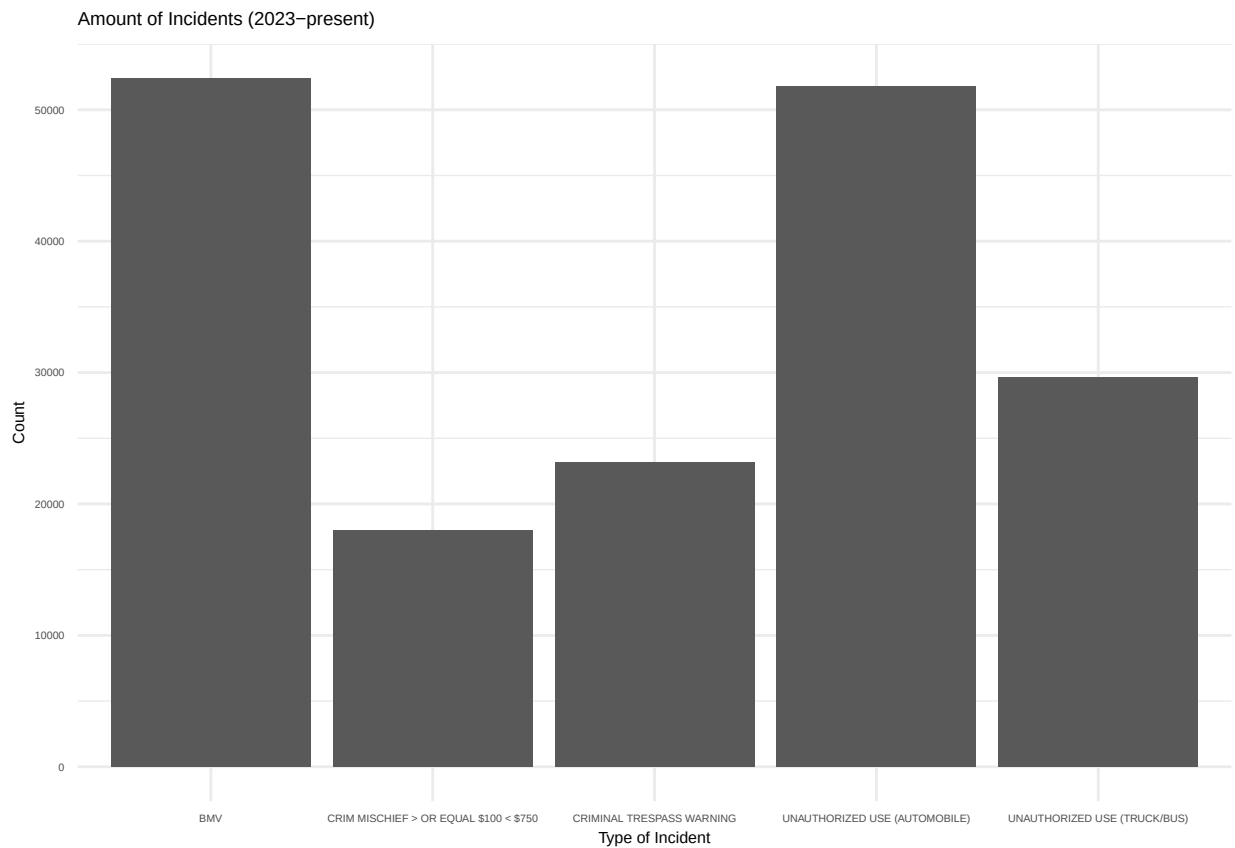
# here we get all the crimes that happened at least 17000 times
dallasCrimes2clean <- dallasCrimes2clean %>% group_by(
  Type.of.Incident) %>% filter(n() > 17000) %>% ungroup()

# for some good data visualization we must change the names of some variables
dallasCrimes2clean$Type.of.Incident[dallasCrimes2clean$Type.of.Incident ==
  "UNAUTHORIZED USE OF MOTOR VEH - AUTOMOBILE"] <-
  "UNAUTHORIZED USE (AUTOMOBILE)"

dallasCrimes2clean$Type.of.Incident[dallasCrimes2clean$Type.of.Incident ==
  "UNAUTHORIZED USE OF MOTOR VEH - TRUCK OR BUS"] <-
  "UNAUTHORIZED USE (TRUCK/BUS)"

# we graph the data again
ggplot(data = dallasCrimes2clean) + geom_bar(aes(x = Type.of.Incident)) +
```

```
theme_minimal() + theme(text = element_text(size = 5.5)) +
labs(title = "Amount of Incidents (2023-present)",
      x = "Type of Incident", y = "Count")
```



Now the data has been cleaned and it is easier to see the most committed crimes so far. We filtered only for the most common ones in order to see which crimes are more likely to be done.